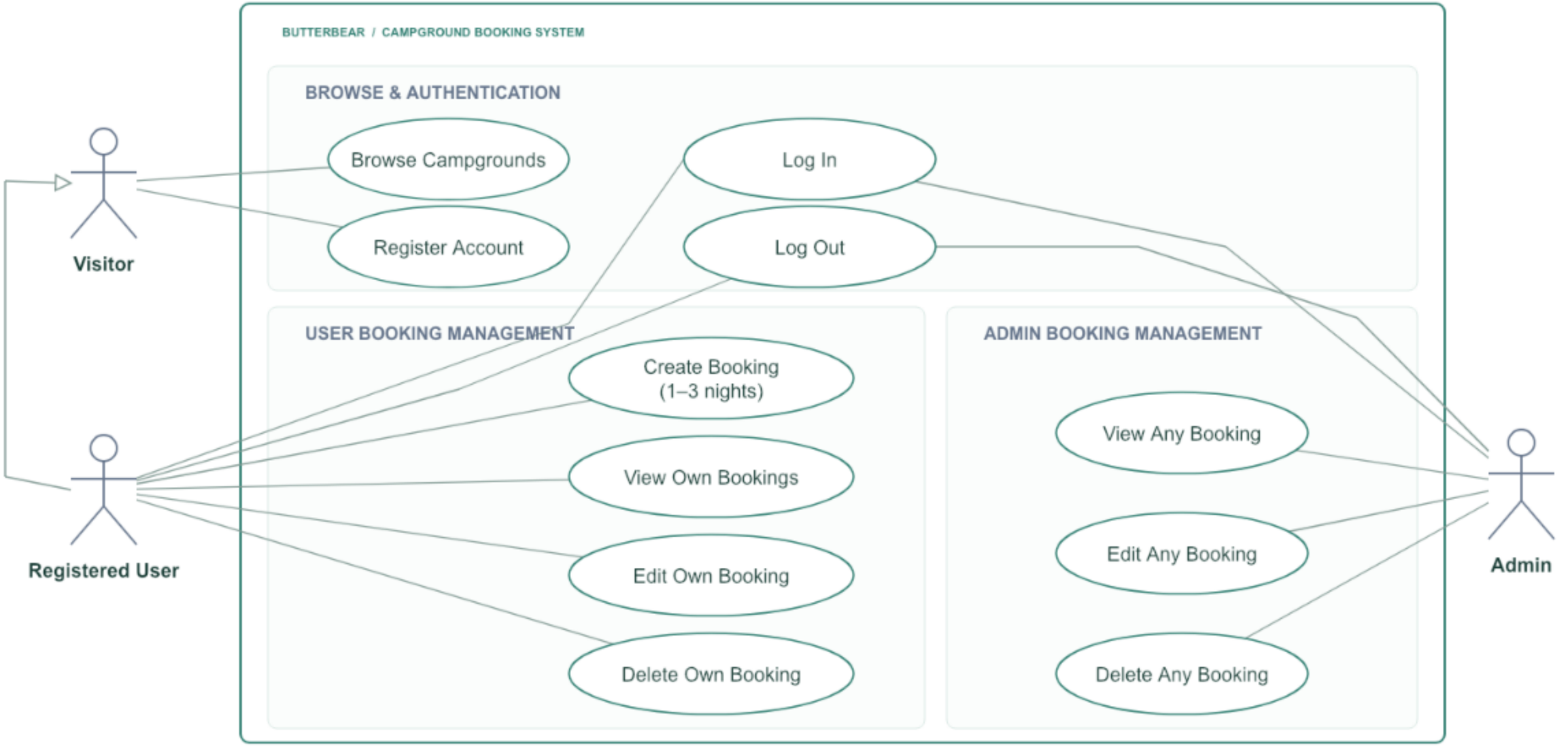
Butterbear | Use Case Diagram

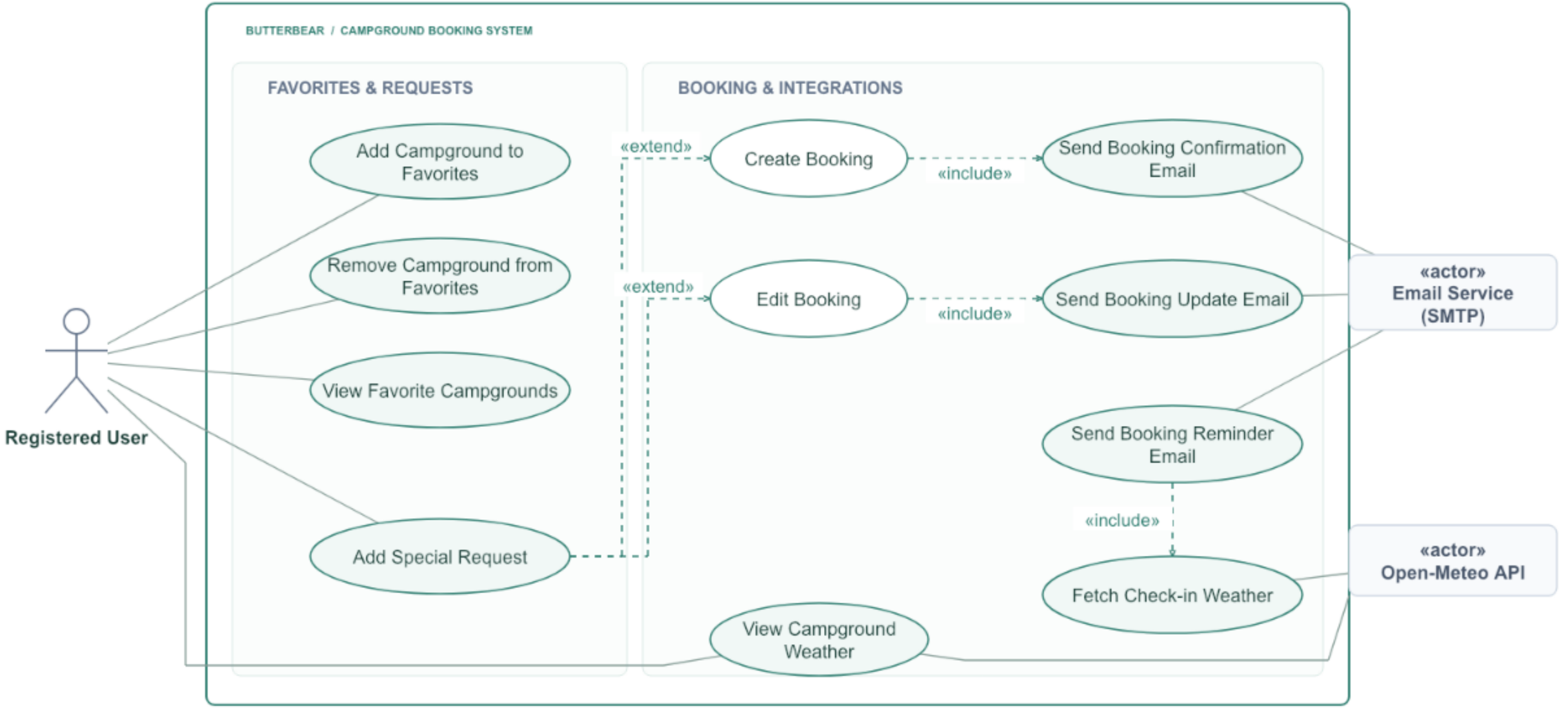
Campground Booking System — Core Functional Overview



Authentication is a precondition for protected actions. Delete Booking cancels the stored booking.

Butterbear | Use Case Diagram

Campground Booking System — Additional Requirements & Integrations



Reminder scheduling runs inside the system. It attempts a weather lookup and sends a clear fallback note if unavailable.

Core API demonstration

USER

01 Register + reject invalid registration

02 Log in

03 View campgrounds

04 Create booking; reject a 4-night stay

05 View own booking

06 Edit own booking

07 Delete own booking

08 Log out; reject the old token

ADMIN

09 Log in as admin

10 View a user booking

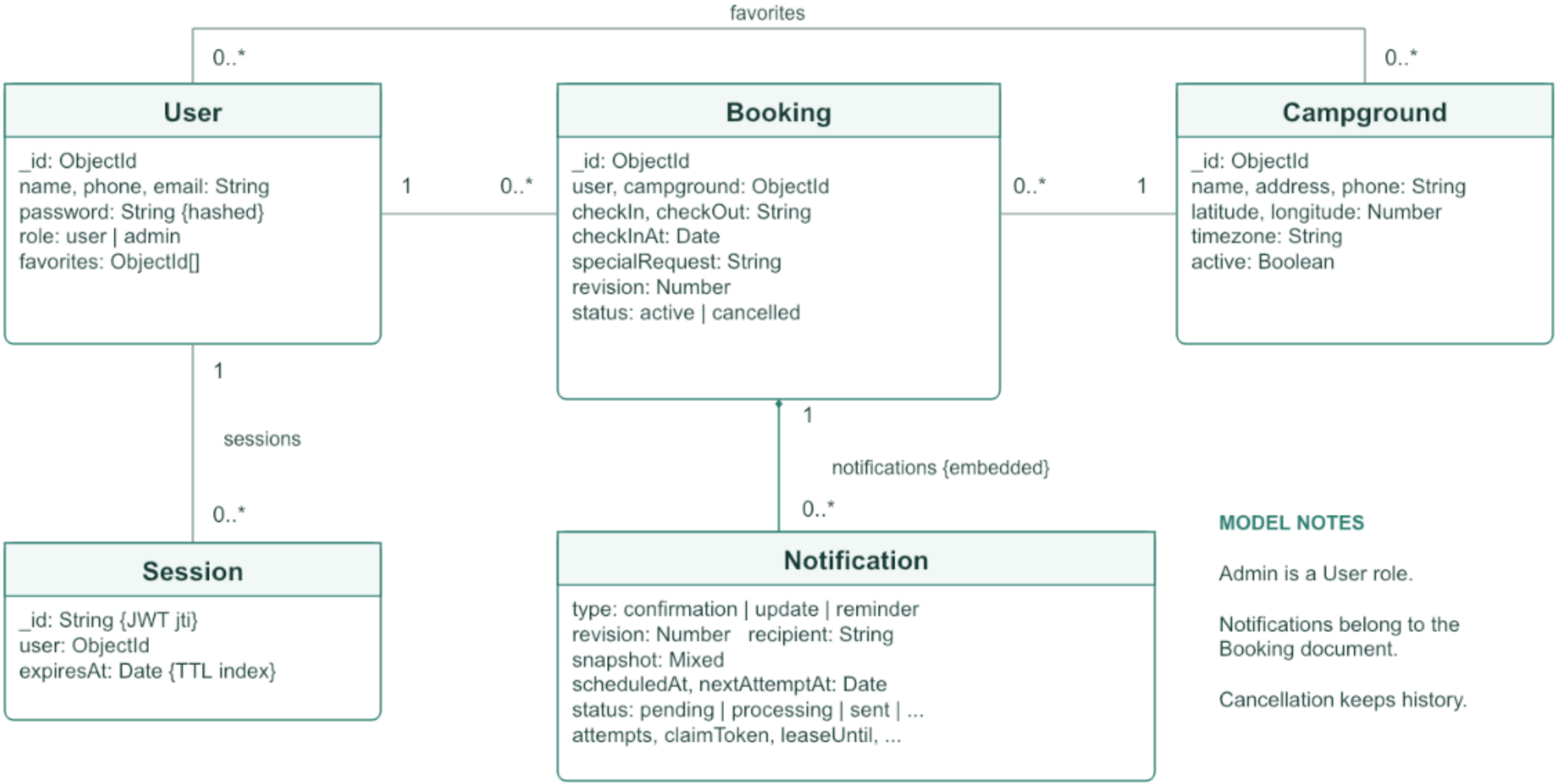
11 Edit that booking

12 Delete that booking

Run folder: 1 Core steps 01-12

Butterbear | Class Diagram

Persistent data model — Mongoose schemas, associations and embedded notifications



MODEL NOTES

Admin is a User role.

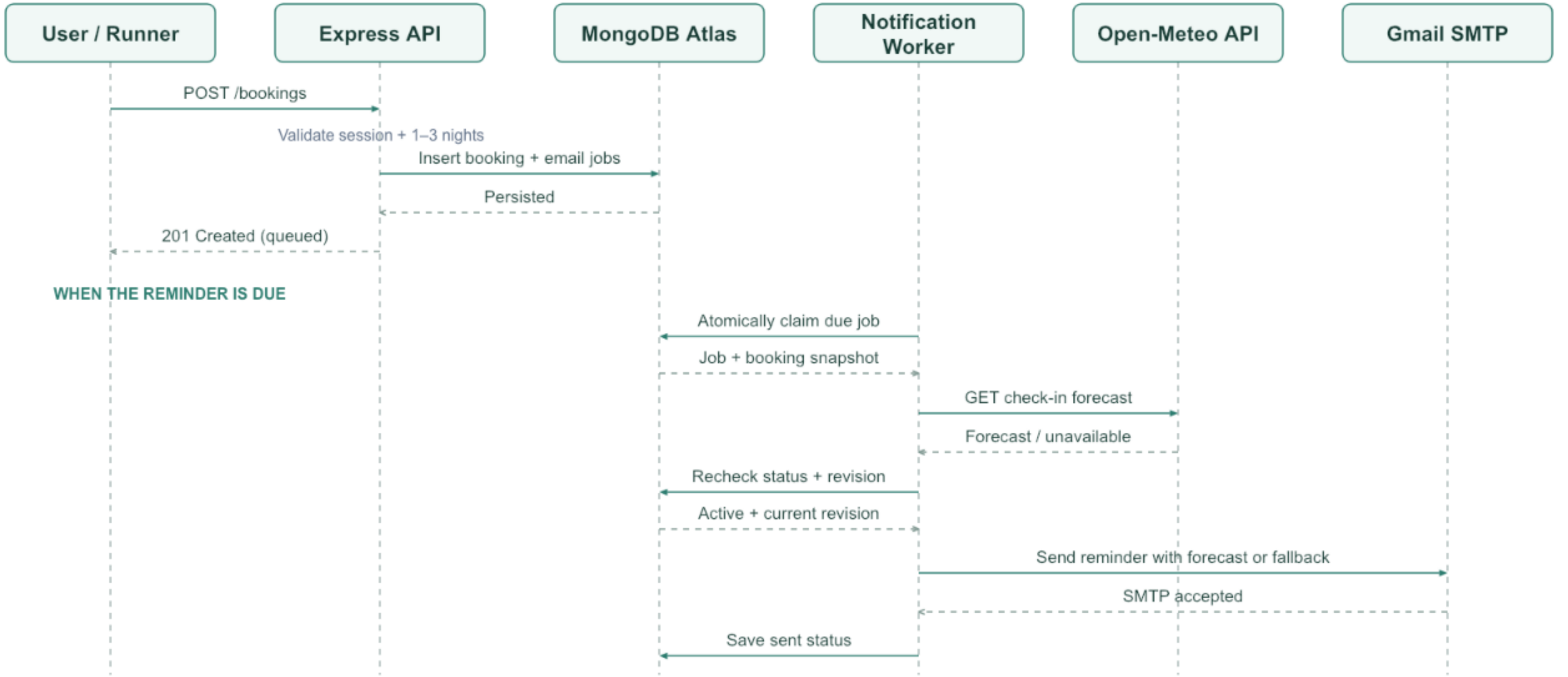
Notifications belong to the Booking document.

Cancellation keeps history.

1–3 calendar nights per booking. Reference relationships use ObjectIds; the filled diamond denotes composition.

Butterbear | Sequence Diagram

Booking creation and a scheduled weather reminder — separate HTTP and background execution



The reminder is normally due at 09:00 the day before check-in. A late booking can make it due immediately.

Extra features | Live demonstration

Favorites	Add → view list → remove
Special request	Save in booking → edit request
Confirmation + update	Show both messages in Gmail
Weather reminder	Open-Meteo forecast inside the email

Run folder: 2 Extra credits step 15

5-minute tutorial video ↗️

Team responsibilities

Nemo

Authentication + session logout

Campgrounds + favorites

Booking creation + date validation

Aon

Booking management + admin scope

Email jobs + retries + SMTP

Weather + reminder scheduling

Equal review packages • 3:30 speaking time each

github.com/chanichkorn/butterbear ↗️

GitHub Insights → Contributors ↗️

AI-assisted implementation; contribution history records actual commits